

Engineering Deterministic Dynamic Setpoint Profiles for Industrial Steam Sterilizers

An advanced Siemens TIA Portal / SCL implementation guide for scan-time-independent, bi-directional thermal ramp control — validated by cyclic-interrupt trace analysis.

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Abstract

Summary

In thermal industrial processes such as pharmaceutical steam sterilization (autoclaves), abrupt changes in temperature setpoints introduce severe thermal stress, mechanical fatigue, and aggressive overshoot in PID loops. This paper details the architecture and implementation of a professional, scan-time-independent dynamic ramp generator developed in Siemens TIA Portal using Structured Control Language (SCL) and executed within a high-precision cyclic interrupt, then validates its heating and cooling response using TIA Portal's watch table and trace tools.

1. The Engineering Challenge: Mitigating Thermal Shock

Steam sterilization requires strict adherence to heating and cooling gradients. Applying a direct step change from room temperature to sterilization temperature — or rapid, unmanaged cooling — forces the PID controller to saturate its output, leading to thermal shock on the vessel walls and unstable jacket pressures.

To solve this, a dynamic setpoint generator (FB_TempRamp_Gen) computes smooth, linear, bi-directional temperature trajectories based on a user-defined ramp time in minutes, rather than leaving the PID loop to chase a step input on its own.

2. Architectural Upgrade: From Cyclic Scan Jitter to Deterministic Execution

Traditional PLC implementations often rely on TON timers executed within the main program cycle (OB1). Because OB1 scan times fluctuate with system load and communication traffic, time-dependent integration performed this way suffers from jitter — the same ramp can run at subtly different rates from one execution to the next.

The Professional Solution

1. **Cyclic Interrupt (OB30):** the ramp function block is isolated inside a dedicated cyclic interrupt configured for a fixed interval ($\Delta t = 100$ ms). This guarantees absolute temporal determinism, independent of what else the CPU is doing that scan.
2. **Continuous Integration Model:** instead of integer truncation steps, the algorithm calculates the exact floating-point position for every cycle, so the trajectory stays mathematically smooth regardless of how long the ramp runs.

$$Output_Value = Start_Temp + \left((Target_Temp - Start_Temp) \div (Ramp_Time_Min \times 60) \right) \times Elapsed_Time$$

3. SCL Core Implementation

The block structure encapsulates its own internal tracking variables — elapsed time, computed rate, and completion flag — so a single FB_TempRamp_Gen can be instantiated once per thermal zone and reused across the plant without any shared state between instances.

```
// 1. Calculate the rate of change in degrees per second
IF #Ramp_Time_Min > 0.0 THEN
    #Ramp_Rate := (#Target_Temp - #Start_Temp) / (#Ramp_Time_Min * 60.0);
ELSE
    #Ramp_Rate := 0.0;
END_IF;

// 2. Execution logic
IF #Reset THEN
    #Elapsed_Time_sec := 0.0;
    #Output_Value     := #Start_Temp;
    #Ramp_Complete    := FALSE;

ELSIF #Enable AND NOT #Ramp_Complete THEN
    // Accumulate elapsed time based on fixed execution interval (dt)
    #Elapsed_Time_sec := #Elapsed_Time_sec + #dt_sec;

    // Calculate dynamic setpoint
    #Output_Value := #Start_Temp + (#Ramp_Rate * #Elapsed_Time_sec);

    // 3. Failsafe clamping and completion check
    IF (#Target_Temp >= #Start_Temp AND #Output_Value >= #Target_Temp) OR
       (#Target_Temp < #Start_Temp AND #Output_Value <= #Target_Temp) OR
       (#Ramp_Time_Min <= 0.0) THEN

        #Output_Value := #Target_Temp;
        #Ramp_Complete := TRUE;
    END_IF;
END_IF;
```

4. Verification, Watch Table Monitoring, and Empirical Trace Analysis

To validate both heating and cooling trajectories prior to live plant deployment, testing was conducted using TIA Portal's runtime monitoring and trace tools, exercising the block instance FB_TempRamp_Gen_DB directly against forced inputs.

A. Heating Phase Analysis (20°C → 50°C)

Watch Table Configuration

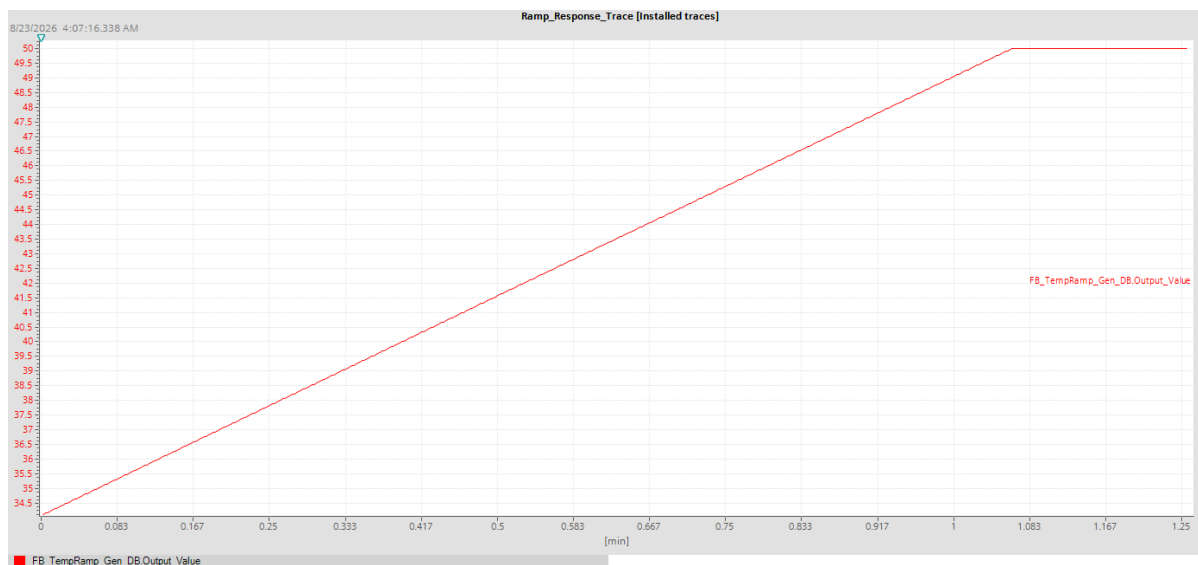
The start temperature was set to 20.0°C, the target to 50.0°C, and the ramp time to 2.0 minutes, with Enable forced TRUE.

Dynamic Setpoint Profiles ▶ PLC_1 [CPU 1212C DC/DC/DC] ▶ Watch and force tables ▶ Watch_Ramp_Test						
	Name	Address	Display format	Monitor value	Modify value	
1	"FB_TempRamp_Gen_DB".Start_Temp		Floating-poin...	20.0	☒	
2	"FB_TempRamp_Gen_DB".Target_Temp		Floating-point nu...	50.0	☒	
3	"FB_TempRamp_Gen_DB".Ramp_Time_Min		Floating-point nu...	2.0	☒	
4	"FB_TempRamp_Gen_DB".Enable		Bool	TRUE	☒	
5	"FB_TempRamp_Gen_DB".Reset		Bool	FALSE	☒	
6	"FB_TempRamp_Gen_DB".Elapsed_Time_sec		Floating-point nu...		☐	
7	"FB_TempRamp_Gen_DB".Output_Value		Floating-point nu...	20.0	☒	
8	"FB_TempRamp_Gen_DB".Ramp_Complete		Bool		☐	
9	<Add new>				☐	

Watch table — heating phase configuration (Start_Temp = 20.0, Target_Temp = 50.0, Ramp_Time_Min = 2.0)

Trace Response

The trace tool confirms a mathematically pure, uniform positive gradient from 20°C to 50°C over the full 2-minute duration, holding flat at the target once Ramp_Complete latches.



Heating trace — Output_Value rising linearly from 20°C to 50°C over 2 minutes, then holding at setpoint

B. Cooling Phase Analysis (80°C → 30°C)

Watch Table Configuration

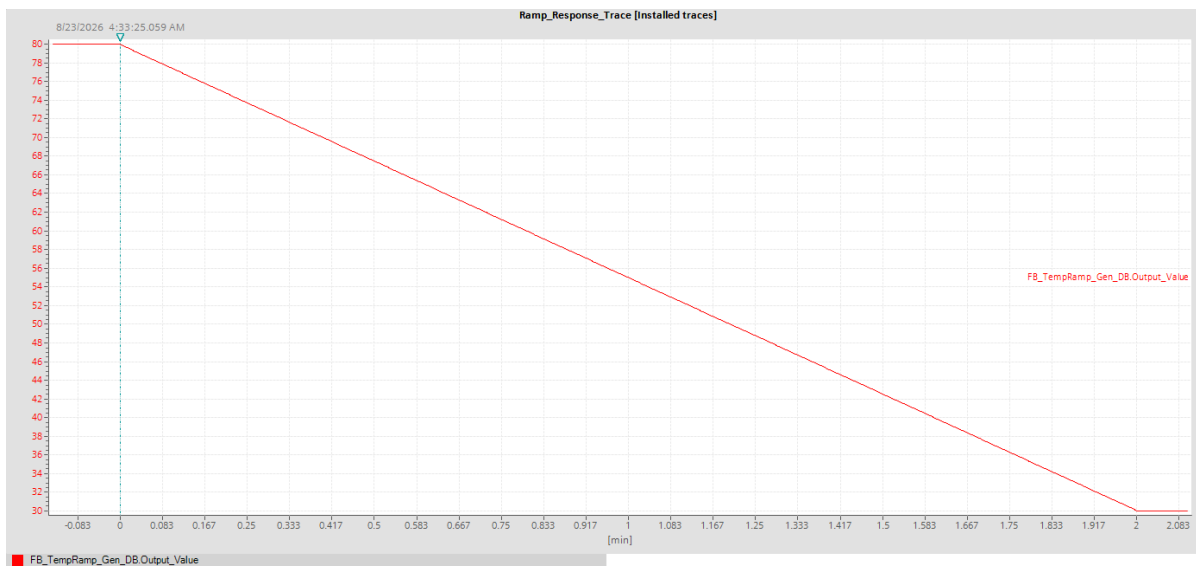
To verify cooling dynamics, the start temperature was configured to 80.0°C and the target temperature to 30.0°C over the same 2.0-minute duration.

Dynamic Setpoint Profiles ▶ PLC_1 [CPU 1212C DC/DC/DC] ▶ Watch and force tables ▶ Watch_Ramp_Test						
	Name	Address	Display format	Monitor value	Modify value	
1	*FB_TempRamp_Gen_DB*.Start_Temp		Floating-point nu...	80.0	80.0	☒
2	*FB_TempRamp_Gen_DB*.Target_Temp		Floating-point nu...	30.0	30.0	☒
3	*FB_TempRamp_Gen_DB*.Ramp_Time_Min		Floating-point nu...	2.0	2.0	☒
4	*FB_TempRamp_Gen_DB*.Enable		Bool	☒ TRUE	TRUE	☒
5	*FB_TempRamp_Gen_DB*.Reset		Bool	☐ FALSE	FALSE	☒
6	*FB_TempRamp_Gen_DB*.Elapsed_Time_sec		Floating-point nu...	120.0987		☒
7	*FB_TempRamp_Gen_DB*.Output_Value		Floating-point nu...	30.0	80.0	☒
8	*FB_TempRamp_Gen_DB*.Ramp_Complete		Bool	☒ TRUE		☐
9	<Add new>					

Watch table — cooling phase, live monitor values mid-ramp (Elapsed_Time_sec = 120.1, Output_Value converging on 30.0)

Trace Response

Because the logic evaluates $\text{Target_Temp} < \text{Start_Temp}$, the block automatically computes a negative Ramp_Rate. The trace output demonstrates a symmetrical, continuous downward gradient, proving that the block handles full bi-directional thermal profiling without deadbands, step jumps, or integral windup.



Cooling trace — Output_Value falling linearly from 80°C to 30°C over 2 minutes

5. Conclusion

This work demonstrates that a scan-time-independent ramp generator, executed inside a fixed-interval cyclic interrupt and driven by a continuous floating-point integration model, eliminates the jitter and step-response risk inherent to OB1-timer-based setpoint changes. Empirical trace analysis confirms mathematically pure, symmetrical behavior in both heating and cooling directions, with clean completion latching and no overshoot

introduced by the setpoint generator itself. The result is a single, reusable SCL block that can be instantiated across every thermally sensitive zone of a sterilization or autoclave process, giving the downstream PID loop a smooth target to track instead of a step it has to fight.

Repository & Documentation

The full FB_TempRamp_Gen SCL source, OB30 cyclic interrupt configuration, and watch table / trace exports are available in the project repository for detailed review.